



AWAIS QURESHI

Applied AI Scientist | Healthcare AI | Bioinformatics | Deep Learning | Generative AI

✉ theawaisqureshi@gmail.com ☎ +92 334 8289446 🔗 00000281892.github.io 🌐 awaisqureshi23

🐙 github.com/00000281892 🆔 orcid.org/0000-0001-7460-3099 📄 scholar.google.com/citations

📍 Islamabad, Pakistan 🌐 Available for Opportunities Across Europe, Saudi Arabia & GCC Countries

PROFESSIONAL SUMMARY

Applied AI Scientist with 14+ years of experience spanning Artificial Intelligence, Software Engineering, Technical Leadership, and Higher Education. Currently completing a PhD in Information Technology at NUST-SEECS, with research focused on Healthcare AI, Bioinformatics, longitudinal microbiome analysis, and explainable deep learning. Published researcher with expertise in Deep Learning, Generative AI, AI Agents, and data-intensive healthcare systems. Committed to developing trustworthy AI solutions that bridge scientific discovery, healthcare innovation, and real-world impact.

EDUCATION

PhD in Information Technology , NUST - SEECS, Islamabad Thesis: <u>Deep Temporal Models for Longitudinal Microbiome Analysis.</u> (Thesis submitted)	01/2019 – Present Islamabad
MS in Information Technology , NUST-SEECS, Islamabad • Thesis: Scalability Analysis of Genome Sequencing with MapReduce in Cloud. • Key Achievement: Awarded AWS Research Credits to deploy and benchmark three Hadoop Clusters on AWS EC2, analyzing genomic data pipeline scalability.	09/2011 – 07/2015 Islamabad

PROFESSIONAL DEVELOPMENT

BUILDING GENAI APPLICATIONS AND AGENTS 📄, COURSERA	07/2026
AWS GENERATIVE AI AND AI AGENTS WITH AMAZON BEDROCK 📄, AWS	07/2026

PROFESSIONAL EXPERIENCE

Kohat University of Science and Technology (KUST)	
Sr. Lecturer Computer Science/FYP Coordinator • Teaching AI, Machine Learning, Software Engineering, and Computing courses while supervising undergraduate final year projects.	10/2020 – 06/2026
Head of the Department • Appointed to lead academic operations, curriculum development, and departmental accreditation compliance.	07/2024 – 10/2025
Bitsym Pvt Ltd, Technical Team Lead • Led development of subscription-based e-learning platform using Moodle. • Integrated local payment gateways (EasyPaiza, JazzCash, MeezanPay).	06/2022 – 05/2026 freelance/remote

Siparadigm Diagnostic Informatics , <i>Senior Software Developer</i> - Developed medical billing and claims management system using ASP.NET MVC. - Improved patient data management and automated claims processing.	06/2019 – 11/2020
Bitsym Pvt Ltd , <i>Senior Software Developer</i> - Built SDN network management system using PHP Laravel. - Enhanced automation and network monitoring capabilities.	08/2018 – 06/2019

TECHNICAL SKILLS

Deep Learning & AI: TensorFlow Keras CNN LSTM Transformers Scikit-learn Big Data & Cloud: Apache Spark Hadoop AWS (EC2, S3) Programming Languages: Python (Advanced) SQL C# Java DevOps & Tools: Docker Git/GitHub Linux/Unix Shell Web & Systems: ASP.NET MVC Laravel RESTful APIs Generative AI & LLMs: Amazon Bedrock AI Agents RAG LangChain Prompt Engineering Hugging Face ChatGPT API	
---	--

PUBLICATIONS & RESEARCH CONTRIBUTIONS

DynaBiome: Interpretable Unsupervised Learning of Gut Microbiome Dysbiosis using Temporal Deep Models. 📄 , <i>BMC Bioinformatics</i> , (Published) Developed an interpretable LSTM-autoencoder framework for longitudinal microbiome dysbiosis detection (ROC-AUC $\approx$ 0.89).	27/02/2026
MetaResNet: An Interpretable Spatial-to-Taxonomic Reverse Mapping Framework for Zero-Shot Microbiome Diagnostics 📄 , <i>BMC Bioinformatics</i> , (Under Review) Designed a CNN-based disease-classification framework for sparse microbiome datasets (AUC > 0.94).	17/05/2026
DynaBiomeX: An Interpretable Dual-Strategy Deep Learning Framework for Architectural Noise Filtration in Sparse Longitudinal Microbiome Data. 📄 , <i>BMC Medical Informatics and Decision Making</i> , (Under Review) Proposed a dual-strategy deep-learning architecture for noise filtration and temporal pattern discovery in longitudinal microbiome data (MCC $\approx$ 0.646).	22/02/2026
CE-ASVID Benchmark Dataset 📄 , <i>GitHub & BMC Bioinformatics</i> Curated and released a longitudinal microbiome benchmark dataset for evaluating temporal deep-learning models in HCT patients.	06/02/2026

ACADEMIC SERVICE & RECOGNITION

Certificate of Appreciation , <i>NUST</i> Organized and conducted the Online Summer School Camp for NUST Entrants Fall 2023 & Fall 2024.	
AWS Research Credits Award , <i>AWS</i> Received AWS research credits for genomics and cloud-based large-scale data processing research.	